

IBRA SATRIATAMA

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Madiun, Jawa Timur, Indonesia

SUMMARY

Electrical Design Engineer with experience in railway electrical system development through projects at PT INKA (Persero). Skilled in electrical schematic design, panel layout development, and 3D cable routing using AutoCAD and Autodesk Inventor for rolling stock applications. Experienced in power electronics, PLC/Arduino-based control systems, and technical engineering documentation. Strong collaborative and problem-solving abilities with attention to detail in structured engineering environments.

WORK EXPERIENCE

PT. Industri Kereta Api (Persero)

Madiun, East Java

Electrical Design Staff

Jan 2026 – Apr 2026

Key Contributions:

- Developed electrical schematics and component layouts for the KRL KCI 8TS project.
- Continued development of electrical panel layouts for the KRL KCI 16TS project using AutoCAD and Autodesk Inventor Electromechanical.
- Created integrated panel schematics and component layouts for the KRL KCI 8TS project.
- Finalized 3D cable routing layouts for the Underframe KRL KCI Retrofit project to optimize space utilization and maintain safe clearances.
- Optimized modular driver desk layouts to improve maintenance accessibility and system flexibility for KRL KCI 8TS.

PT Inka Multi Solusi Consulting

Madiun, East Java

Electrical Design Staff

Oct 2025 – Dec 2025

Key Contributions:

- Continued development of electrical schematics and component layouts for the KRL KCI Retrofit project.
- Continued development of electrical panel layouts for the KRL KCI 16TS project using Autodesk Inventor.
- Designed 3D cable routing layouts for the Underframe KRL KCI Retrofit project to optimize space utilization and maintain safe clearances.
- Prepared and verified Bill of Materials (BOM) and component specifications for electrical panel assemblies.
- Performed cable sizing calculations and prepared cable schedules for power distribution systems.
- Managed technical drawing revisions and engineering documentation using Microsoft Excel.

INTERNSHIP EXPERIENCE

Neosia Training Center (Industrial Mechanical & Product Design)

Remote

MSIB Batch 7 – Independent Study

Sep 2024 – Dec 2024

Key Contributions:

- Contributed to the design and presentation of a hydraulic press machine for industrial waste processing as a final engineering project.
- Developed 50+ 2D and 3D engineering designs through intensive CAD/CAM/CAE training using SolidWorks, AutoCAD Mechanical, and Autodesk Inventor.
- Produced manufacturing drawings and mechanical models to support fabrication and assembly processes.
- Completed the Certified SolidWorks Professional (CSWP) international certification assessment to validate technical design competency.

PT. Industri Kereta Api (Persero)

Madiun, East Java

Electrical Design Intern

Jan 2024 – Jun 2024

Key Contributions:

- Developed electrical schematics and designed electrical panel layouts for railway systems including braking, auxiliary power, lighting, and control systems using CAD software.
- Created 3D component and assembly models using Autodesk Inventor to support visualization, prototyping, and manufacturing processes.
- Designed cable routing systems and prepared technical documentation including cable schedules, wiring diagrams, and electrical specifications for train manufacturing and maintenance activities.

EDUCATION

Madiun State Polytechnic

Aug 2021 – Aug 2025

D4 Rollingstock Engineering (Electrical Concentration)

GPA: 3.67 / 4.00

Thesis : Design and Development of a PID-Control-Based Battery Management System with a Buck-Boost Converter for Battery Performance Optimization.

PROJECTS

Battery Management System (BMS) with Buck–Boost Converter & PID Control

- A Battery Management System (BMS) integrated with a PID-controlled Buck–Boost Converter, designed to maintain stable output voltage across buck and boost modes, evaluated under open-loop and closed-loop control for electric vehicle and energy storage applications.
- Tools: Arduino IDE, EasyEDA, Matlab, Oscilloscope, Multimeter.

SEPIC Converter as Input for 3-Phase Inverter (PI Control)

- Growing demand for renewable energy has driven extensive research in power conversion. Conventional step-up DC converters, however, remain limited in output voltage range and efficiency. DC-DC converter technology addresses this gap by enabling flexible voltage conversion across DC levels, making high step-up capability with maintained efficiency critical for electric vehicle and renewable energy applications.
- Tools: Arduino IDE, EasyEDA, Matlab, Oscilloscope, Multimeter.

Hydraulic Press Machine for Industrial Waste Processing

- Designed and tested a hydraulic press machine including structural analysis and full manufacturing drawings using SolidWorks and AutoCAD Mechanical.
- Tools: AutoCAD, Autodesk Inventor

CERTIFICATIONS

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|---|-----------------|
| • Certified SolidWorks Professional (CSWP) – Sheet Metal Dassault Systèmes | Jan 2025 |
| • AutoCAD, Inventor & Vault 2025 Training PT INKA (Persero) | Oct 2025 |
| • SolidWorks, AutoCAD Mechanical, Autodesk Inventor Neosia Training Centre | Dec 2024 |

ORGANIZATIONAL EXPERIENCE

Rollingstock Engineering Student Association

Jan 2023 – Jun 2024

Division of Information & Communication

- Designed visual materials for internal and external communication needs, including posters, banners, social media feeds, and infographics.
- Created informative and visually engaging content to support publication of organizational events and campaigns.

SKILLS

Software Proficiency :

Microsoft 365, Autodesk Vault, AutoCAD, Autodesk Inventor, SolidWorks, PLC Omron/Siemens, Arduino, Programming (C/C++, Python)

Soft Skills :

Analytics, Leadership, Teamwork, Critical Thinking, Effective Communication, Problem Solving, Time Management, Adaptability, Collaborative.

Languages :

Indonesian: Native | **English:** Intermediate – TOEFL ITP 533